THE UNIVERSITY OF MELBOURNE

Student number

Semester 1 Assignment 3, 2025

School of Mathematics and Statistics

MAST20009 Vector Calculus

Submission deadline: 5pm on Monday May 5

This assignment consists of 10 pages (including this page) with 3 questions and 37 total marks

Instructions to Students

- This assignment is worth 5% of your total MAST20009 mark.
- If you have a printer, print the assignment one-sided.

Writing

- Assignments must be neatly written in blue or black pen or may be typeset using LaTeX.
- Full working must be shown in your solutions.
- Marks will be deducted for incomplete working, insufficient justification of steps, incorrect mathematical notation and for messy presentation of solutions.
- If you are writing answers on the assignment and you need more space, use blank paper. Note this in the answer box, so the marker knows.
- If you are only writing on blank A4 paper, the first page must contain only your student number, subject code and subject name. Write on one side of each sheet only. Start each question on a new page and include the question number at the top of each page.

Scanning and Submitting

- Put the pages in number order and the correct way up. Add any extra pages to the end. Use a scanning app to scan all pages to PDF. Scan directly from above. Crop pages to A4.
- Submit your scanned assignment as a single PDF file and carefully review the submission in Gradescope. Scan again and resubmit if necessary.

Page 1 of 10 — add any extra pages after page 10 — Page 1 of 10

Page 2 of 10 — add any extra pages after page 10 — Page 2 of 10

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Page 3 of 10 — add any extra pages after page 10 — Page 3 of 10

Question 1 (10 marks)

Let $\mathbf{F}$ be the vector field in $\mathbb{R}^3$ given by

$$\mathbf{F}(x, y, z) = (xz + xy)\,\mathbf{i} + \alpha(yz - xy)\,\mathbf{j} + \beta(yz + xz)\,\mathbf{k}$$

where $\alpha, \beta \in \mathbb{R}$ are constants. Suppose that $\mathbf{F}$ satisfies $\mathbf{F} = \nabla \times \mathbf{G}$ for some smooth vector field $\mathbf{G}$.

(a) Find α and β.

By Identity 9, we have

$$\nabla \cdot \mathbf{F} = \nabla \cdot (\nabla \times \mathbf{G}) = 0.$$

Then

$$\begin{aligned} 0 = \nabla \cdot \mathbf{F} &= \nabla \cdot (xz + xy, \alpha(yz - xy), \beta(yz + xz)) \\ &= (z + y) + \alpha(z - x) + \beta(y + x) \\ &= x(-\alpha + \beta) + y(1 + \beta) + z(1 + \alpha). \end{aligned}$$

Hence we must have $\alpha = \beta = -1$.

Page 4 of 10 — add any extra pages after page 10 — Page 4 of 10

(b) Now suppose that

$$\mathbf{G}(x,y,z) = xyz\,\mathbf{i} - xyz\,\mathbf{j} + g(x,y,z)\,\mathbf{k}$$

for some function g. Find $g(x,y,z)$.

We have

$$\nabla \times \mathbf{G} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ xyz & -xyz & g(x,y,z) \end{vmatrix}$$
$$= (g_y + xy)\,\mathbf{i} - (g_x - xy)\,\mathbf{j} + (-yz - xz)\,\mathbf{k}$$

Comparing this to

$$\mathbf{F} = (xz + xy)\,\mathbf{i} - (yz - xy)\,\mathbf{j} - (yz + xz)\,\mathbf{k}$$

we must have

$$g_y = xz$$
$$g_x = yz$$

Integrating the first equation with respect to y gives

$$g = xyz + h(x,z).$$

Differentiating this with respect to x gives $g_x = yz + h_x$, and hence $h_x = 0$. So in fact $h(x,z) = \ell(z)$ is independent of x.

We finally conclude that $g(x,y,z) = xyz + \ell(z)$ for some function $\ell(z)$, which is independent of x and y.

Page 5 of 10 — add any extra pages after page 10 — Page 5 of 10

Question 2 (16 marks)

(a) Consider the curve C made up of two straight line segments, from $P = (0,0,0)$ to $Q = (1,2,2)$, and then from Q to $R = (4,2,-2)$. Compute the integral

$$\int_C (xz + y)\, ds.$$

We must parametrise the two segments separately. First let C_1 be the segment PQ, for which one parametrisation is

$$\mathbf{c}_1(t) = (0,0,0) + t(1,2,2) = (t, 2t, 2t), \quad 0 \le t \le 1.$$

On C_1 we have

$$\frac{ds}{dt} = \left|\frac{d\mathbf{c}_1}{dt}\right| = \sqrt{1^2 + 2^2 + 2^2} = 3.$$

So

$$\begin{aligned}\int_{C_1} (xz + y)\, ds &= \int_0^1 ((t)(2t) + 2t) \cdot 3\, dt \\ &= 3 \int_0^1 (2t^2 + 2t)\, dt \\ &= 3 \left[\frac{2t^3}{3} + t^2\right]_{t=0}^{t=1} \\ &= 5.\end{aligned}$$

Next we parametrise $C_2 = QR$, using

$$\mathbf{c}_2(t) = (1,2,2) + t(3,0,-4) = (1 + 3t, 2, 2 - 4t), \quad 0 \le t \le 1.$$

Then

$$\frac{ds}{dt} = \left|\frac{d\mathbf{c}_2}{dt}\right| = \sqrt{3^2 + 0^2 + (-4)^2} = 5.$$

So

$$\begin{aligned}\int_{C_2} (xz + y)\, ds &= \int_0^1 ((1 + 3t)(2 - 4t) + 2) \cdot 5\, dt \\ &= 5 \int_0^1 (-12t^2 + 2t + 4)\, dt \\ &= 5 \left[-4t^3 + t^2 + 4t\right]_{t=0}^{t=1} \\ &= 5.\end{aligned}$$

Page 6 of 10 — add any extra pages after page 10 — Page 6 of 10

Combining,

$$\int_C (xz + y)\, ds = \int_{C_1} + \int_{C_2} = 10$$

Page 7 of 10 — add any extra pages after page 10 — Page 7 of 10

(b) Let C be the circle of radius 2 with centre $(1,1)$, and let C_1 be the curve along one quarter of C, from $(3,1)$ to $(1,3)$ in the anticlockwise direction. Define the force field $\mathbf{F}(x,y) = (2x, x-y)$. Find the work done by $\mathbf{F}$ in moving a particle along C_1.

We must first parametrise C_1. One possible parametrisation is

$$\mathbf{c}_1(t) = (1 + 2\cos t, 1 + 2\sin t), \quad 0 \le t \le \tfrac{\pi}{2}$$

Then then work done by $\mathbf{F}$ is

$$\begin{aligned}
\int_{C_1} \mathbf{F} \cdot d\mathbf{c} &= \int_0^{\frac{\pi}{2}} (2 + 4\cos t, 2\cos t - 2\sin t) \cdot (-2\sin t, 2\cos t)\, dt \\
&= \int_0^{\frac{\pi}{2}} (4\cos^2 t - 4\sin t - 12\cos t \sin t)\, dt \\
&= \int_0^{\frac{\pi}{2}} (2 + 2\cos 2t - 4\sin t - 6\sin 2t)\, dt \\
&= [2t + \sin 2t + 4\cos t + 3\cos 2t]_{t=0}^{t=\frac{\pi}{2}} \\
&= (\pi - 3) - (4 + 3) \\
&= \pi - 10.
\end{aligned}$$

Page 8 of 10 — add any extra pages after page 10 — Page 8 of 10

Question 3 (11 marks)

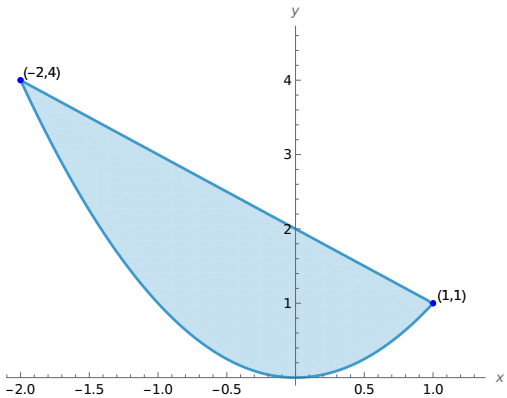
Let D be the region in $\mathbb{R}^2$ between the parabola $y = x^2$ and the line $y = 2 - x$. Let $f(x, y) = \frac{x}{y}$.

(a) Consider the integral

$$I = \iint_D f(x, y)\, dA.$$

Write down an expression for I as a double integral with *vertical* strips. You should include a sketch of the region D, labelling the relevant features. You *do not* need to evaluate the integral.

The region D looks like this:



The intersection point on the left is $(-2, 4)$ and the one on the right is $(1, 1)$. We then have

$$\iint_D f(x, y)\, dA = \int_{-2}^{1} \int_{x^2}^{2-x} \frac{x}{y}\, dy\, dx$$

(b) Now compute the integral I, this time using *horizontal* strips.

To use horizontal strips we must split the region D into two parts: D_1 for $0 \le y \le 1$ and D_2 for $1 \le y \le 4$. In D_1 we then have $-\sqrt{y} \le x \le \sqrt{y}$, while in D_2 we have $-\sqrt{y} \le x \le 2 - y$.

Firstly for D_1,

$$\begin{aligned}
\iint_{D_1} f(x,y)\,dA &= \int_0^1 \int_{-\sqrt{y}}^{\sqrt{y}} \frac{x}{y}\,dx\,dy \\
&= \int_0^1 \left[\frac{x^2}{2y}\right]_{x=-\sqrt{y}}^{x=\sqrt{y}} dy \\
&= 0.
\end{aligned}$$

Then for D_2,

$$\begin{aligned}
\iint_{D_2} f(x,y)\,dA &= \int_1^4 \int_{-\sqrt{y}}^{2-y} \frac{x}{y}\,dx\,dy \\
&= \int_1^4 \left[\frac{x^2}{2y}\right]_{x=-\sqrt{y}}^{x=2-y} dy \\
&= \int_1^4 \left(\frac{(2-y)^2}{2y} - \frac{1}{2}\right) dy \\
&= \int_1^4 \left(\frac{2}{y} - \frac{5}{2} + \frac{y}{2}\right) dy \\
&= \left[2\log y - \frac{5y}{2} + \frac{y^2}{4}\right]_1^4 \\
&= -\frac{15}{4} + 4\log 2.
\end{aligned}$$

Page 9 of 10 — add any extra pages after page 10 — Page 9 of 10

Page 10 of 10 — add any extra pages after page 10 — Page 10 of 10

End of Assignment — Total Available Marks = 37